ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. AI-DS ENGINEERING — SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain 5 V's of Big Data: Volume, Velocity, Variety, Veracity, and Value. [14 Marks]

[OR]

Q.1 (Alt) Compare RDBMS vs NoSQL databases: Document, Key-Value, Column-Family, and Graph stores. [14 Marks]

--- UNIT II ---

Q.2 Explain Apache Hadoop Ecosystem: HDFS NameNode, DataNode architecture and Secondary NameNode. [14 Marks]

[OR]

Q.2 (Alt) Explain YARN (Yet Another Resource Negotiator) architecture: ResourceManager and NodeManager. [14 Marks]

--- UNIT III ---

Q.3 Explain Apache Spark Architecture: RDDs (Resilient Distributed Datasets), Transformations, and Actions. [14 Marks]

[OR]

Q.3 (Alt) Compare Spark in-memory computation vs Hadoop MapReduce disk-based execution. [14 Marks]

--- UNIT IV ---

Q.4 Explain CAP Theorem (Consistency, Availability, Partition Tolerance) with real database examples. [14 Marks]

[OR]

Q.4 (Alt) Describe Apache Cassandra architecture: Gossip protocol, Consistent Hashing, and SSTables. [14 Marks]

--- UNIT V ---

Q.5 Explain Stream Processing with Apache Kafka: Topics, Partitions, Producers, and Consumer Groups. [14 Marks]

[OR]

Q.5 (Alt) Describe Big Data pipeline for sentiment analysis using Spark MLlib and GraphX. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****